

Hydra MAGE — Candidate-Generation Supplement

Companion Supplement — Novel Superconductor Candidate Generation

Companion to *Hydra MAGE: A 2048-Parameter Effective Theory of Superconductivity*

*Draft. Documents the generative-discovery run behind main report §4.9. Every candidate below is a **hypothesis**, not a **discovery**: no synthesis or experimental validation has been performed, and the model’s Tc head is not an independent oracle. Read §3 and §6 before treating any number as physical.*

1 Purpose & Scope

This supplement records, in full, the candidate-generation run that the main report references in one line. It covers **how** generation was done (§2), **how that compares to how generative models are judged in other domains** (§3), the **candidate catalogue and its honest funnel** (§4), an **in-depth look at the most interesting candidates** (§5), and a **failure/honesty ledger** (§6). Reproducibility details are in §7.

- **Model:** the 13563 successor (adds a physical-validity / possibility head), of near-identical latent geometry to the paper’s interpretability model (§4.8 of the main report).
- **Decoder:** all candidates were decoded **exclusively with the two-stage decoder** (predicts the element set, then the fractions, in single passes — no autoregressive drift), the same clean path used for holdout recovery.

2 How Generation Occurred — Algorithms

2.1 Setup

All strategies operate in the 2048-dimensional latent space; a candidate is a latent vector $\mathbf{z}$ decoded to a formula and scored by the model’s own heads. Seeds are drawn from a 74,765-material latent cache built for the 13563 model (§7).

2.2 The three strategies (focused sweep)

Strategy	Idea	Mechanism
S0 — cheap-metal $\rightarrow$ SC	turn an inexpensive non-superconductor into a candidate	gradient ascent on P(SC) from a metal seed, then on Tc, regularized against the Magpie manifold ($R^2=0.86$) to stay physically coherent; seeds located by composition-similarity in the cache
S4 — cross-family interpolation	blend two families’ latent regions	spherical interpolation (SLERP) between diversified seed pairs across all C(13,2)=78 family pairs — <i>this is the strategy that produced the flagship Hf-FeSe</i>
S8 — Tc-targeted	request a target Tc	gradient ascent from random latent points toward a conditioned Tc

Strategies S1–S3, S6, S7 (dome-peak, λ -directed, gradient-opt, element-substitution, Sobol) exist in the pipeline but were **excluded** from this focused run to keep the candidate set interpretable.

2.3 Filtering & scoring pipeline (cheap deterministic checks first)

0. **Basic-chemistry / physics validity gate — deterministic, run *before* any database or scoring compute.** A rule-based filter, *not* a model head (the learned validity head is overconfident off-manifold — it scored the nonsense $\text{Ts}_{0.5}\text{N}_{0.5}$ at 0.97). It **hard-rejects**: unsynthesizable elements (superheavy $Z \geq 99$ incl. all transactinides, plus At/Fr), inert noble-gas constituents (He/Ne/Ar, which form no bulk compound), absurd stoichiometry (element count outside 1–10, non-positive fractions), and **family-inconsistent decoder artifacts** (a claimed cuprate with no Cu or O, an iron-based with no Fe, an MgB_2 -type with no B). It **soft-flags** (keeps, de-prioritizes): all-metallic compositions, ≥ 2 toxic heavy metals, radioactive-but-real elements, heavy noble gases (Kr/Xe/Rn), and two **physics flags** on predicted Tc — extraordinarily high in absolute terms (> 200 K), and **above the realistic ambient ceiling for the claimed family** (a conventional/BCS metal above ~ 40 K, an iron-based above ~ 60 K, a cuprate above ~ 140 K). The family-ceiling flag is what removes the “conventional metal at 110 K” artifacts that pure Tc- or economics-ranking would otherwise promote.

*On SMACT charge-neutrality — evaluated and deliberately **not used as a filter**.* We installed SMACT (the standard charge-balance / electronegativity screen) and tested it on the superconductor domain. It is **inappropriate here**: its integer-oxidation-state model mis-rejects nearly every real SC family — verified failures on $\text{YBa}_2\text{Cu}_3\text{O}_7$, doped $\text{La}_{1.85}\text{Sr}_{0.15}\text{CuO}_4$, MgB_2 , H_3S , and doped iron-based SCs — because the mixed-valence hole/electron doping that *defines* high-Tc superconductivity cannot be represented by integer oxidation states, and because it cannot parse fractional doped stoichiometries. Using it as a filter would discard the best candidates. It remains available as an **opt-in advisory** (`smact_charge_check()`) for the one case where it is valid: simple, clean-stoichiometry ionic compositions (e.g. a generated binary oxide) *outside* the cuprate/boride/hydride/metallic families — where it correctly accepts Na_2O and rejects the impossible K_2O_3 .

1. **6-head validation** — P(SC), Tc, family, Magpie-manifold consistency, element-count, and the 13563 validity/possibility head.
2. **Cross-decoder agreement** — AR vs two-stage decoding disagreement flags unreliable generations (a reference-free out-of-distribution signal).
3. **Two-stage novelty check** — (1) against the training corpus, then (2) against **six databases** (**$\sim 365\text{k}$ entries**): training, NEMAD SC/magnetic/thermoelectric/anisotropy, and Materials Project, matched on canonical composition and element set.
4. **Economic scoring** — a composite 0–100 (raw-material cost, cooling tier, processing complexity, supply risk, application match). *Reported as context, **not used to select or rank the showcase candidates** — see the note at the end of §5.*

3 How This Compares to Generative SOTA in Other Domains

Generation is judged very differently across fields; positioning this run against them shows which axes we cover and which we honestly do not.

3.1 The cross-modal evaluation matrix

Axis		Images / Video	LLMs / Text / Audio	Molecules / Materials	Protein / Code / Tabular	Our composition+Tc analog
Fidelity (on-manifold?)		Density; precision	FAD/KAD; BERTScore	Validity (SMACT); FCD	scRMSD; MMD	validity/possibility head; Magpie-manifold
Diversity / coverage		Coverage; recall	distinct-n; Vendi	Uniqueness; COV	COV; 1-NNA	unique formulas / Magpie Density–Coverage
Distribution-match		FID → CMMD	MAUVE; KAD	FCD; property-EMD	MMD; 1-NNA	kernel-MMD on element+Tc histograms
Novelty vs memorization		NN-copy detect	extraction / MIA	novelty vs ref DB; Authenticity	DCR + MIA	element-set vs stoichiometry novelty (differ $\sim 10\times$)
Conditional success		VQAScore	CLAP; pass@k	S.U.N., DFT-verified	pass@k; scRMSD	joint composition+Tc (our V.N.T. ; defined §3.3)
Independent oracle		human eval	MOS / human	DFT; BEE-NET	wet-lab; execution	physics.tc Δ ; cross-decoder disagreement
Downstream utility		—	—	screening pipelines	TSTR	(not yet run)

3.2 The load-bearing principle: *don’t let the model grade its own homework*

The single strongest cross-domain convergence is that the verifier of conditional success must be **independent of the generator**: proteins refold with a separate predictor (scRMSD $< 2 \text{ \AA}$), code executes against unit tests (pass@k, and *hardening the tests re-ranks models* — EvalPlus), materials verify with DFT rather than the generator’s own property head (MatterGen), and image text-faithfulness moved from self-echoing CLIPScore to independent VQAScore.

Our honest position: the Tc head *is* the generator, so it is **not** an independent oracle. We ground it with (a) the **physics-formula Tc** (Allen-Dynes / Presland) as a Δ signal, (b) **cross-decoder disagreement**, and (c) the **validity head** — but a fully independent Tc oracle (e.g. BEE-NET via structure inversion) is **not yet in the loop**. This is the central limitation of every Tc number below.

3.3 S.U.N. $\rightarrow$ our V.N.T., as a conjunction

Materials generation converged on **S.U.N.** (Stable, Unique, Novel, reported *simultaneously*) because marginals mislead — 100% valid $\wedge$ 100% novel $\wedge$ 100% on-target can each hold on disjoint subsets while the intersection is ≈ 0 . Our analog is **V.N.T.** — **Valid** $\wedge$ **Novel** $\wedge$ **Tc-in-target** — reported as the intersection in §4.

3.4 Novelty at two granularities

As in the LLM literature (n-gram novelty, extraction/MIA) and image NN-copy detection, novelty is not one number. We report **element-set novelty** (a genuinely new combination of elements) separately from **stoichiometry novelty** (a new ratio of a known element set) — these differ by roughly $10\times$ in our audits, and conflating them inflates “novelty.”

3.5 The gameability warning (AddCarbon)

A trivial “insert one random atom” baseline can score Validity ≈ 1.0 / Novelty ≈ 1.0 and beat real generative models on naive metrics (Renz 2019). We therefore treat distribution-match (hardest to fool) as the discriminating axis and flag, in §6, exactly where our “novelty” is a one-atom perturbation of a known material.

3.6 Quality–diversity is a Pareto frontier

Aggressive property optimization (our S0/S8 gradient ascent, high steering) is the regime where “alignment trades diversity for target-hitting $\rightarrow$ mode collapse.” The S8 candidates in §4 are a live example: conditioning on ~ 110 K collapsed a large fraction of them onto one cuprate-doping cluster.

3.7 Scorecard for this run

Axis	Covered here	How
Fidelity	✓	validity head + Magpie-manifold + reliability filter
Diversity/coverage	⊙	unique-formula count; full Density/Coverage deferred to the atlas run
Distribution-match	✗	not computed (recommended next)
Novelty (2 granularities)	✓	6-database check, element-set vs stoichiometry
Conditional success (V.N.T.)	⊙	Valid $\wedge$ Novel $\wedge$ Tc-target — but Tc not independently verified
Independent oracle	✗	the key gap — Tc head is the generator
Downstream utility (TSTR)	✗	future work

4 Candidate Catalogue — Overview

4.1 The funnel

The three **bolded** rows are the actual sequential funnel (raw $\rightarrow$ gate $\rightarrow$ working pool). The un-bolded per-filter rows are **independent marginal pass-counts**, each applied to the full 1,683-candidate raw set — they do not nest or narrow monotonically; only their intersection (the working pool) does.

Stage	Count
Raw candidates generated (S4 1,547 · S8 117 · S0 19)	1,683
Passed basic-chemistry / physics validity gate (deterministic, pre-search)	1,407
— hard-rejected by the gate	276 (271 family-inconsistent · 4 unsynthesizable · 1 stoichiometry)
<i>Per-filter pass-counts (each independently, of the 1,683 raw):</i>	
— unique two-stage formulas	1,682
— reliability ≥ 0.7	1,431
— flagged valid	1,184
— novel vs training (in-pipeline)	1,315
Working pool = novel $\wedge$ reliable $\wedge$ not-collapsed $\wedge$ Tc$\in$[10,160] K (intersection)	171
— of pool: element-set NOVEL (deep 6-database check)	73
— of pool: novel-stoichiometry (RELATED/VARIANT)	98
Distinct element sets in the pool	97

4.2 Distributions

- **Strategy:** 92% S4 (interpolation), 7% S8 (Tc-target), 1% S0 (cheap-metal). S0 yields few survivors — pushing a metal into the SC region rarely produces a *novel* composition.
- **Tc (reliable pool):** min 0.6 K, median **40.5 K**, max 313.2 K. The high tail is dominated by S8 conditioning at ~ 110 K and a small number of implausibly-high BCS/hydride outliers.
- **Magnetic heads:** across the entire pool, **0 candidates** tripped the ferromagnet (`fm_prob>0.5`) or antiferromagnet (`afm_prob>0.5`) heads — i.e. no magnetic-transition false positives of the kind documented in the main report contaminated this pool. This is a genuine positive of decoding on-manifold via the two-stage decoder.
- **Two-stage collapse:** 1,397 / 1,683 raw candidates carried a `ts_collapsed` flag — a reminder that most sampled latent points do *not* decode to coherent novel materials, and that the filters are doing real work.

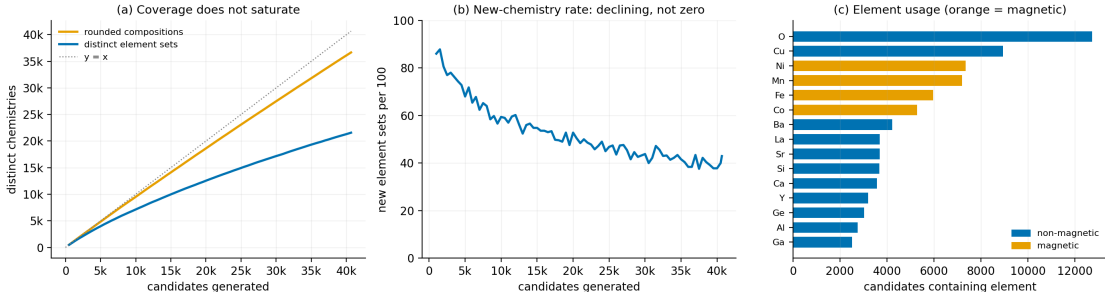
4.3 The honest headline

Element-set “novelty” is dominated by trace-dopant technicalities and S8 conditioning artifacts, not new chemistry. The most chemically plausible candidates are a smaller set of **iron-based** compositions from S4 cross-family interpolation (§5, iron-based group) — though the §5 literature capstone shows even these resolve to known-family variants, i.e. **zero clean novel survivors**. The high-Tc (~ 110 K) “cuprate” cluster is largely a conditioning + decoder artifact (§5, high-Tc cuprate cluster; §6).

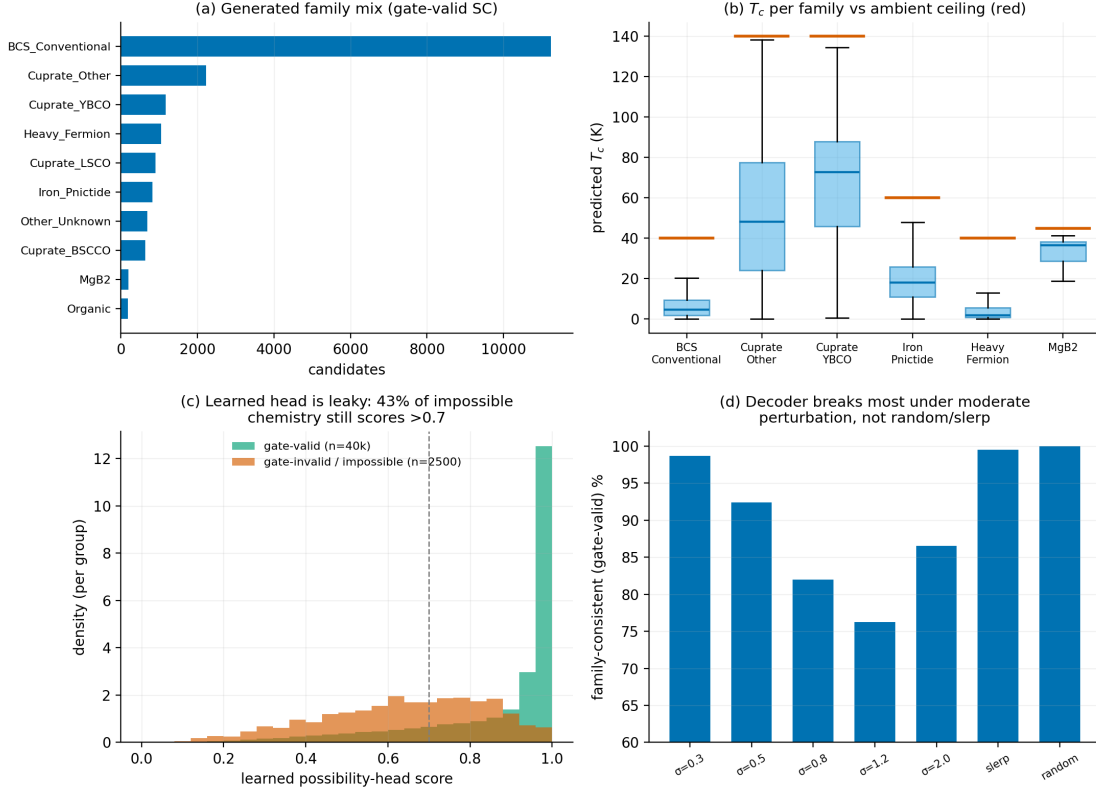
4.4 Coverage atlas — a 43,000-candidate generation run

Beyond the focused S0/S4/S8 run, a **43,264-candidate coverage atlas** was generated by *pure forward sampling* of the latent space — Gaussian perturbation of training seeds ($\sigma = 0.3\text{--}2.0$), spherical interpolation (SLERP) between seed pairs, and unbiased random-z — each decoded with the two-stage decoder, annotated with the **full trained-head suite**, and passed through the deterministic validity gate (94% pass). The purpose is *characterization*, not discovery: how broad is the model’s producible chemistry, and how do its heads behave at scale? (*Reproducibility: `atlas_sampler.py`, `atlas_analysis.py`, `head_relationships.py`, `z_analysis.py`; data `atlas_4h/atlas_4h.csv`, 43,264 rows $\times$ full heads. The five figures below are also bundled as `generation_atlas_figures.pdf`.)*

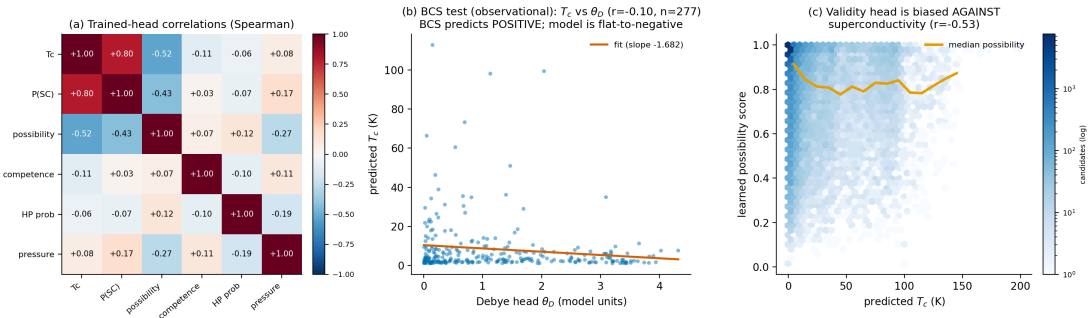
Coverage does not saturate. The 40,670 gate-valid candidates span **21,578 distinct element sets** (36,665 rounded compositions), still climbing at the end of the run: the new-element-set rate declines from 86 to 41 per 100 candidates but never plateaus, and 8,862 element-sets appear only in the second half. The model’s *producible* chemical space is thus far larger than 100k candidates would exhaust — but this is coverage of what the model can *emit* (basic-validity-filtered), **not of plausible superconductors**: filtering to sensible, novel candidates collapses it by $\sim 1000\times$ (§5). The **vast-producible / narrow-plausible** gap is the honest characterization of the generative manifold.



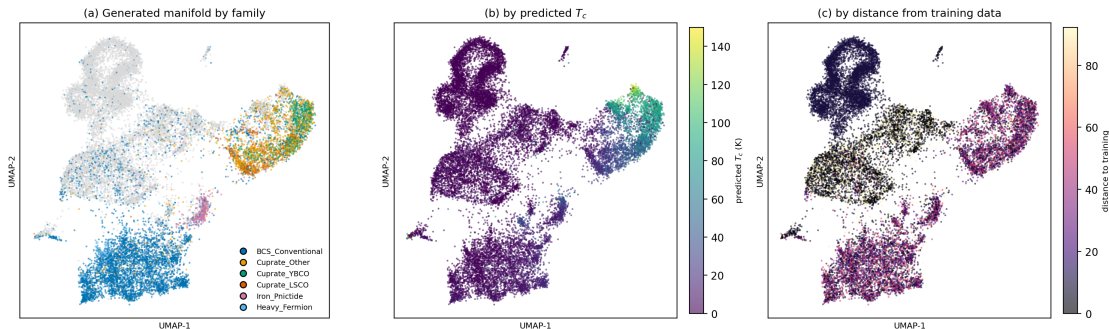
Family mix, Tc ceilings, and where the filters bite. Generated families are dominated by a low-Tc BCS_Conventional bucket (median 4.5 K), then the cuprates; predicted Tc largely **respects family-appropriate ceilings** (cuprates spread to ~ 140 K, iron-based below ~ 60 K, MgB_2 near 40 K). The gate hard-rejects 6% — almost all **family-inconsistent decoder collapse** (a “cuprate” with no Cu/O), which peaks under *moderate* perturbation ($\sigma \approx 1.2 \rightarrow 76\%$ valid) and is negligible for small perturbations, interpolation, and random-z.



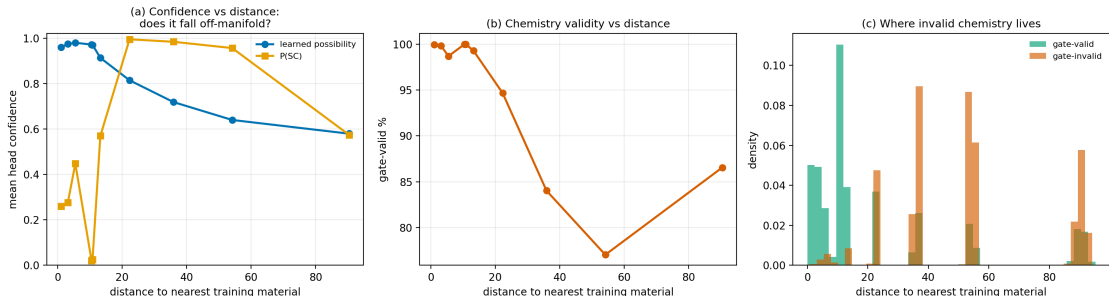
Head relationships expose two calibration facts. (i) The Debye and Tc heads are flat-to-negatively correlated (observational, $n=277$) — corroborating the causal $\beta \approx 0$ result that the model’s Tc is **not phonon-stiffness-driven**. (ii) The learned possibility head is **biased against superconductivity** ($r = -0.53$ with Tc, -0.43 with $P(\text{SC})$): it rates high-Tc/SC candidates as *less* plausible, favouring mundane compositions — another reason not to use it as a filter.



The generated manifold is structured. A UMAP of the generated latents is **family-organized** (distinct BCS, cuprate, iron, and heavy-fermion regions), with high predicted Tc localized to the **cuprate archipelago**, and a large **off-manifold region** far from any training material.



Off-manifold overconfidence is a *dissociation* — the atlas’s strongest result. Binning by distance to training, the SC classifier’s $P(SC)$ rises to ~ 1.0 off-manifold (near training it is correctly low, tracking the $\sim 57\%$ non-SC neighbourhood), while the **possibility head falls from 0.96 to 0.58** — genuine out-of-distribution awareness, and the reason it was added to the successor. The two heads *disagree* off-manifold, and that disagreement is a usable, reference-free signal. The possibility head is nonetheless **leaky** (43% of gate-impossible chemistry still scores >0.7), which is why the pipeline gates deterministically rather than trusting the learned score. (*This result is referenced in the main report §“Confident Failures Off the Manifold”.*)



Two honest corrections from this analysis. (1) The pool shows **zero magnetic false positives** — but *not* because the magnetic heads catch them: those heads are **entirely inactive** in generation (all null). The real success is the SC/ T_c head correctly down-rating magnetic-element soups (24% of candidates are Mn/Cr/Fe-rich; median $T_c \approx 0$ K). (2) The possibility head is **not** blanket-overconfident — it is the more OOD-aware head (it falls with distance); the SC head is the overconfident one. A refinement of the coarse “the head is overconfident” reading.

5 Candidate Showcase — Sensible, Not-Obviously-Wrong Chemistry

What this section is for (read first). This is *not* a ranked discovery list and *not* an economic screen. Its purpose is to show that the model — filtered only by the deterministic validity gate (§2.3) and novelty — produces **chemically plausible, family-consistent candidates across every superconductor mechanism**: sensible doped/substituted variants of real families, at family-appropriate T_c , not gibberish. Every entry below **passes the validity gate, is novel vs the six databases, sits under its family’s ambient T_c ceiling, and did not trip the magnetic heads**; one exemplar is shown per distinct chemistry, ordered by the validity head within each mechanism. All-metallic *multi-toxic* “soups” with no compound-forming rationale (e.g. Ca-Cd-Hg-Pb, §6.1) are **excluded** from the showcase but retained in the full atlas; toxic-but-*real* families (e.g. Tl/Hg cuprates, as in real HBCCO/TBCCO) are kept and noted. **poss** = validity/possibility

head; econ = economic composite (reported, **not** the selection criterion — see the note at the end). Source: `shortlist_clean.csv` (gate-applied; note that its `db_classification` column predates the training-corpus recheck — K-La-H and Nb-In-Cu are training reproductions despite their CSV labels, per the §5 literature capstone). Formulas are the two-stage decoder’s output.

Iron-based (doped pnictides / oxypnictides) — the most chemically sensible group

Iron-based superconductivity is induced by transition-metal doping of a parent 122/1111 compound — exactly what these are. Same S_4 lineage as the flagship Hf-FeSe.

Formula	Tc (K)	poss	Novelty	Rationale
Rb_{0.15}(Fe,Co)As	24	0.95	element-set novel	Co-doped Rb-122 — <i>the</i> textbook route to SC in 122 pnictides. Most sensible hit.
Ba_{0.2}(Fe,Co)As	13	0.97	stoichiometry	Co-doped Ba-122, the canonical Ba(Fe,Co) ₂ As ₂ family.
Ba_{0.2}(Fe,Ni)As	8.5	0.99	—	Ni-doped Ba-122 — the other standard 122 doping route.
Sr-Sc-Fe-P-O	16	0.94	stoichiometry	an iron oxypnictide (1111-type, LaFeAsO family) with Sc/P — coherent.
Rb-Fe-Co-As-Se / Eu-La-Fe-As	10–17	0.95	mixed	anion- or rare-earth-substituted iron-based; plausible, lower confidence.

MgB₂-type (dopant substitutions on the 39 K boride)

Formula	Tc (K)	poss	Rationale
Mg_{0.33}Sc-B	42	0.99	Sc-modified MgB ₂ near the parent Tc; speculative but coherent.
Mg-Sn-B	29	0.98	Sn substitution — plausible carrier-doping variant.
Mg-Al-B	18	0.99	Al-doped MgB₂ is real and <i>lowers</i> Tc — the model reproduces the right direction.

Cuprate (plausible-Tc, family-consistent)

Formula	Tc (K)	poss	Rationale
Hg-Ba-Cu-O	43	0.997	Hg-cuprate motif (HBCCO is the real highest-Tc family); clean Cu–O.
Tl-Hg-Ba-Ca-Cu-O	124	0.99	multi-cation Tl/Hg cuprate — real highest-Tc chemistry (toxic, as the real ones are; soft-flagged).
Bi-Sr-Ca-Cu-O	12	0.995	BSCCO-type; Bi-cuprate.

Metallic / alloy (the class you flagged as commercially interesting)

Low-Tc but real: Nb–Ti/Nb₃Sn are the workhorse SC magnets. A novel metallic SC above today’s operating temperatures would be commercially significant — see the economics note.

Formula	Tc (K)	poss	Rationale
Zr-Ti-Nb-Si	11	0.99	Nb-Ti-Zr refractory alloy (Nb-Ti is the workhorse); plausible intermetallic.
Nb-In-Cu	9	0.99	Nb-based intermetallic. <i>Present near-verbatim in the training corpus → reproduction, not novel; Tc≈elemental Nb.</i>
Ca-Al-Ga-Si, Mg-Al-Si	5–6	0.99	light-metal intermetallics; modest Tc.

Hydride & other

Formula	Tc (K)	poss	Rationale
K-La-H	48	1.00	hydride; LaH_x superconducts at high Tc but only at $\sim 150\text{--}200$ GPa — treat as a high-pressure hypothesis (§6). <i>Present in the training corpus (K-La hydride) $\rightarrow$ reproduction, not novel.</i>
Al-B-C	9	0.997	borocarbide-like light-element compound.
Eu-Fe-S / Ca-Y-Tm-O	10–13	0.99	Eu-Fe-S carries a magnetic-element caution (Eu); rare-earth oxide otherwise.

Flagship carryovers (prior work, reconciled with this run)

- **$\text{Fe}_{0.9}\text{Hf}_{0.09}\text{Se}$** (~ 21 K) and **$\text{BaFe}_2(\text{As,Se})_2$** (~ 26 K) — the main report’s candidates; the iron-based group above are their natural neighbours, independently resurfaced by the 13563 run.

On economics — a deliberately separate, more nuanced question

An economic composite (0–100) is reported per candidate but is **not** the selection criterion here, for two reasons. First, this section is about chemical *plausibility*, not commercial viability. Second, the composite favours **cheap, ambient, common-element** materials (it ranks the MgB_2 variants highest) — but that is not where commercial value necessarily lies: a **novel metallic/alloy superconductor with a Tc above today’s operating materials** ($\text{Nb-Ti} \sim 9$ K, $\text{Nb}_3\text{Sn} \sim 18$ K, $\text{MgB}_2 \sim 39$ K) could be far more valuable than a marginally-cheaper low-Tc one, even though the composite would score it lower. Commercial assessment is its own analysis; here it is context, not a ranker.

Literature verification of the showcase — the honest capstone

The showcase candidates cleared the deterministic gate and an automated novelty screen (two entries, noted below, are reclassified as training reproductions on a direct training-corpus check), but only the flagship Hf-FeSe had had a manual literature search. We closed that gap with a documented web literature search over **all 22** (three query tiers — exact composition, parent + dopant, near-neighbour — plus a known-magnet cross-check for magnetic-element compositions; full per-candidate table with references in `literature_search.md`, 2026-07-16). The result is decisive and honest:

- **10/22 are exact or near rediscoveries of known superconductors** — a near-exact match to the real **17 K oxypnictide $\text{Sr}_4\text{Sc}_2\text{Fe}_2\text{P}_2\text{O}_6$** (predicted 16.5 K), textbook optimal-doped **$\text{Ba}(\text{Fe,Co/Ni})_2\text{As}_2$** , near-pure MgB_2 and Al-doped MgB_2 , **CaAlSi** , and the **Hg-1201 / La-214 / Tl-Hg-1223** cuprates. This *validates the model’s compositional recall* across every major family.
- **6/22 are RELATED** — a known SC family whose exact doping was not found (Rb-(Fe,Co)As, K-La-H, Ti-Nb-Zr-Si, La-Ni-Si, a Sr-deficient BSCCO, a La-doped EuFe_2As_2).
- **3/22 are known non-superconductors** (false positives) — most tellingly **Nb-In-Cu**, where Nb and Cu are immiscible and the predicted 9.1 K $\approx$ elemental Nb’s 9.26 K (an *elemental-Nb echo*); plus Al-B-C ceramics and an insulating $(\text{Y,Tm})_2\text{O}_3$ sesquioxide.
- **1/22 is a phase-transition confusion** — **Eu-Fe-S**, whose predicted “Tc” sits in the Eu^{2+} magnetic-ordering band (the documented failure mode of §6).
- **2/22 are NOVEL only because they are chemically implausible artifacts** — Fe-Ni-As-Se-H (a mixed-anion + hydrogen phase that does not exist) and Mg-Sn-B (Sn does not enter the MgB_2 lattice).

- **Two are also training reproductions.** A direct canonical-composition check against the training corpus places **K-La-H** and **Nb-In-Cu** (the latter near-verbatim) *inside* the training set — so beyond their literature verdicts (RELATED and known-non-SC) they are reproductions, not novelties. Only **six** of the 22 are genuinely absent from the training corpus (Rb-(Fe,Co)As, Zr-Ti-Nb-Si, Al-B-C, the (Y,Tm)-oxide, Eu-Fe-S, Fe-Ni-As-Se-H), and — as the verdicts above show — none of those six is a confirmed superconductor.
- **Genuinely-novel and plausible survivors: zero.** Even the flagship Hf-FeSe, while unpublished, carries a real objection — Fe-site substitution is documented to *suppress* FeSe superconductivity (the directly analogous Ti-doped FeSe is reported non-superconducting).

The lesson is the paper’s thesis at its sharpest: **database-novel $\neq$ literature-novel $\neq$ plausible**. The generation loop’s value is a *disciplined recall / interpolation* of known superconducting chemistry, not a fountain of new materials; the gate $\rightarrow$ novelty $\rightarrow$ literature pipeline is exactly what separates the one flagged hypothesis from the rediscoveries and the artifacts. (*Honesty caveat: a web search is not exhaustive — “NOVEL” here means “no published work found in the searched sources, 2026-07-16.”*)

6 Failure Modes & Honesty Ledger

0. **The corrected pipeline order (the lesson that reorganized this run).** The failures below are *why* a deterministic basic-chemistry / physics validity gate now runs **first**, before any database search or scoring (§2.3). The three that follow are exactly what it catches — cheaply, before spending compute.
1. **Off-manifold overconfidence, and a reproduced-nonsense trap — the clearest failure.** $\text{Ts}_{0.5}\text{N}_{0.5}$ is not a creative error: tennessine ($Z=117$) appears in exactly **5 of the 74,765 seed materials**, all deliberate vocabulary-expansion gibberish, and this is one of them — the S4 interpolation landed off-manifold and the decoder’s barely-trained Ts slot re-emitted an *injected-nonsense training string*. It then exposed two blind spots: the *learned* validity head scored it **0.97** (the P(SC)/validity overconfidence the main report documents — a high head score is not evidence of a real material), and the 6-database novelty check flagged it **NOVEL** — trivially, since tennessine compounds are in no real database, so **the novelty check cannot distinguish genuinely-new chemistry from injected nonsense reproduced**. Only the *rule-based* gate catches it (unsynthesizable element). A contrasting case — **Ca-Cd-Hg-Pb**, which the gate keeps but soft-flags because it is *implausible* rather than impossible — is worked through in §6.1.
2. **Decoder element-dropping.** Several high- T_c “cuprates” are missing Cu or O — the two-stage decoder, pushed to a conditioned Tc, produces a cuprate-*shaped* fingerprint but omits a defining element (Sr-Ca-C-O is really a carbonate; Ba-Ca-Cu-Hg has no oxygen). These are now caught deterministically by the gate’s **family-essential-element** check (271 of the 276 hard rejections were of this kind).
3. **Conditioning mode-collapse.** S8’s ~ 110 K target produced a large near-duplicate cuprate cluster. Element-set “novelty” via trace dopants (Fe, S, C, Cl, Ni added to a cuprate base) is a **one-atom-perturbation** technicality — exactly the AddCarbon gameability failure (§3.5), not new chemistry.
4. **Hydride pressure blindness.** The model has no structure or pressure input for the K-La-H hydride candidate; a 48 K ambient hydride prediction ignores the ~ 150 – 200 GPa that real LaH_x superconductivity requires.
5. **Magnetic false positives — absent here (a positive).** Unlike the main report’s magnetic-transition contamination, no candidate in this pool tripped the FM/AFM heads. Decoding on-manifold via the two-stage decoder, plus the validity head, kept the pool clear of the Curie/Néel-scale confusions.

6.1 Worked example — “flagged, not forbidden”: Ca-Cd-Hg-Pb

The gate keeps $\text{Ca}_{0.164}\text{Cd}_{0.574}\text{Hg}_{0.139}\text{Pb}_{0.594}$ (soft flags: all-metallic + multiple toxic heavy metals) rather than hard-rejecting it, and that restraint is deliberate — it illustrates the honest line between *implausible* and *impossible*. It should **not** be in the showcase, but the reasoning is worth recording.

Artifact tell. The decoder’s fractions sum to **1.47**, **not 1** — they do not normalize, a decoder-artifact signature rather than a real composition.

Why it is an implausible superconductor (not why it is “impossible”):

- **No driver for a defined quaternary compound.** Ca is strongly electropositive ($\chi \approx 1.0$) and forms Zintl-type phases with p-block metals (Ca–Pb: Ca_2Pb , CaPb_3 ; Ca–Hg amalgams are real), but Cd, Hg and Pb are mutually miscible soft metals with small electronegativity differences that form **disordered amalgams/eutectics, not fixed-stoichiometry phases** (assessed diagrams in Massalski, *Binary Alloy Phase Diagrams*, ASM 1990). The system would most plausibly **phase-separate**, not crystallize as the single ordered phase a superconductor needs.
- **Low-Tc constituents, no enhancement motif.** Cd 0.52 K, Hg 4.15 K (Kamerlingh Onnes 1911), Pb 7.2 K (compiled in Roberts, *J. Phys. Chem. Ref. Data* **5**, 581, 1976); Ca is the highest-Tc *elemental* SC but only under ~ 200 GPa ($T_c \approx 29$ K at 216 GPa, Sakata et al., *Phys. Rev. B* **83**, 220512, 2011) and is **not** superconducting at ambient. A random alloy of soft, heavy, low-Debye-temperature sp-metals is a low-coupling, low-DOS BCS system (Bardeen–Cooper–Schrieffer, *Phys. Rev.* **108**, 1175, 1957) with none of the Tc-raising motifs (Al₅, Chevrel, MgB_2 σ -bands, CuO_2 planes), and Matthias’s valence-electron rules (*Phys. Rev.* **97**, 74, 1955) place high-valence heavy sp-metal mixtures away from favourable Tc — so the model’s **14 K** prediction sits above all four constituents with no mechanism to justify it.
- **Toxicity** (Cd/Hg/Pb) is a real but *practical* objection — correctly a soft flag, not a rejection.

Where the reasoning honestly stops. None of this proves impossibility: a metastable Ca–Cd–Hg–Pb inter-metallic could conceivably be quenched into existence, and alloying occasionally raises Tc above constituents. Settling it needs the independent-oracle tier this supplement flags as not-yet-run — a **DFT convex-hull / phonon** check for whether any ordered phase is even stable, then an Allen–Dynes Tc estimate. Until then it is *implausible*, which is why it lives here in the ledger rather than in §5. It is the clean case where the **validity gate correctly declines to call it impossible while the validity/possibility head wrongly calls it confident (poss $\approx$ 0.99)** — the honest verdict lives in the gap between them.

What would falsify the iron-based candidates: a DFT/MLIP relaxation showing no stable structure at the predicted composition; a BEE-NET (structure-based) Tc far below the head’s prediction; or a literature entry we missed. None has been run — these are the honest next steps.

7 Reproducibility

- **Model:** outputs/checkpoint_final_13563.pt (epoch 13563).
- **Seed cache:** outputs/latent_cache_13563_full.pt (built by scratch/gen13563/build_13563_full_cache.py).
- **Command:** `GEN_CHECKPOINT_PATH=outputs/checkpoint_final_13563.pt GEN_ZCACHE_PATH=outputs/latent_cache_13563_full.pt PYTHONPATH=srcpythonscripts/generation/generation_pipeline_v2.py --strategies 0,4,8 --skip-mp`
- **Head annotation (full trained-head suite per candidate):** scratch/gen13563/annotate_heads.py $\rightarrow$ outputs/generation_v2/candidate_head_atlas_13563.{csv,parquet,_z.pt} ($1,683 \times 37$ fields).

- **Deep novelty triage:** `scratch/gen13563/triage.py` (6-database NoveltyVerifier); diversification `scratch/gen13563/diversify.py`.
- **Validity gate:** `basic_chemistry_validity()` in `generation_pipeline.py` (standalone audit + tuning: `scratch/gen13563/validity_gate.py`), applied in `post_process_candidates` **before** novelty and economic scoring; annotated atlas with gate results in `scratch/gen13563/atlas_with_gate.csv`.
- **Pipeline port changes:** `GEN_CHECKPOINT_PATH` / `GEN_ZCACHE_PATH` / `GEN_ZCACHE_OUTPUT` env overrides added to `generation_pipeline.py` and `rebuild_z_cache.py`; `load_latent_cache` accepts a pre-built formulas key; `_resolve_full_z_cache` honours the override.

Appendix A — Full candidate table

`outputs/generation_v2/candidate_head_atlas_13563.csv` (all 1,683 candidates, every trained-head prediction + generation metadata). Ranked shortlists: `scratch/gen13563/{triaged_candidates,shortlist_final}.csv`.

Appendix B — Metric glossary (one line each)

Density/Coverage (Naeem 2020, outlier-robust precision/recall) · CMMD (CLIP+MMD distribution distance) · MAUVE (text distribution divergence) · FCD (ChemNet Fréchet distance) · S.U.N. (Stable-Unique-Novel conjunction) · scRMSD (protein self-consistency RMSD) · pass@k (code execution success) · TSTR (train-synthetic-test-real) · Authenticity (Alaa 2022, per-sample generalization).